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Body fat gain and loss differentially influence validity of dual-energy x-ray absorptiometry and multifrequency bioelectrical impedance analysis during simultaneous fat-free mass accretion

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ABSTRACT

The validity of dual-energy x-ray absorptiometry (DXA) and multifrequency bioelectrical impedance analysis (MFBIA) for detecting changes in fat mass (FM), fat-free mass (FFM), and body fat percentage (BF%) was evaluated, as compared to a rapid 4-component (4C) model, in 31 females completing 8 weeks of resistance training. Analyses were performed in all participants (ALL) and in subgroups that gained FFM but lost FM (R subgroup) or gained both FFM and FM (G subgroup). It was hypothesized that methods would comparably detect changes in ALL, but discrepancies would occur in subgroup analysis. Changes in body composition did not significantly differ between 4C, DXA, and MFBIA. Equivalence testing indicated that similar changes were detected by DXA and MFBIA, compared to 4C, for ΔFFM in all analyses and ΔBF% in ALL and R subgroup. ΔFM was equivalent to 4C only in R subgroup for DXA and G subgroup for MFBIA. For ΔFM and ΔBF%, DXA and MFBIA produced similar magnitude errors in ALL. However, DXA exhibited lower error in R subgroup, whereas MFBIA exhibited lower error in G subgroup. For ΔFFM, DXA and MFBIA exhibited relatively similar errors in ALL and R subgroup, although MFBIA displayed proportional bias and weaker correlations with 4C than DXA. In G subgroup, MFBIA exhibited lower errors and a higher correlation with 4C ΔFFM than DXA. Although both DXA and MFBIA may have utility for estimating body composition changes during FFM accretion, DXA may be superior during simultaneous FM loss, whereas MFBIA may produce lower error during simultaneous FM gain.

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Abbreviation: 4C, 4-component model; ANOVA, analysis of variance; BF%, body fat percentage; BIA, bioelectrical impedance analysis; BIS, bioimpedance spectroscopy; BM, body mass; BMC, bone mineral content; BV, body volume; CE, constant error; DXA, dual-energy x-ray absorptiometry; FFM, fat-free mass; FM, fat mass; G, subgroup of participants who gained both fat-free mass and fat mass; LOA, 95% limits of agreement; LST, lean soft tissue; MFBIA, multifrequency bioelectrical impedance analysis; Mo, bone mineral; R, subgroup of participants who gained fat-free mass and lost fat mass; RT, resistance training; SEE, standard error of the estimate; TBW, total body water; TE, total error.

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1. Introduction

The relationship between body composition and health, disease, and athletic performance is widely recognized as a topic of interest to health professionals and practitioners [1]. However, the validity of commonly used body composition assessment methods varies widely [1,2]. While a large number of studies have reported the relative agreement between methods, these investigations are typically cross-sectional in nature. Although establishing the error associated with a method at a single point in time provides some useful information, the ability of a particular method to accurately track *changes* in body composition over time is arguably more important because it allows for the quantification of responses to a lifestyle intervention or disease process. Even if some amount of absolute disagreement between criterion and alternate methods is observed cross-sectionally, the ability to detect changes in body composition may establish a method as appropriate for longitudinal use.

Dual-energy x-ray absorptiometry (DXA) and bioelectrical impedance analysis (BIA) are widely used for the estimation of body composition in clinical and research settings [3-5]. However, shortcomings of both methods have been reported when compared to multicomponent models, such as the molecular-level 4-component (4C) model that divides body mass into fat, water, bone mineral, and residual components [6-8]. Some of the limitations of DXA and BIA may be due to the assumption of fat-free mass (FFM) characteristics, such as the water, protein, and mineral content of FFM, as intra- and interindividual variations in these properties have been reported [9-11]. Because of these concerns, it has been suggested that a multicomponent model, which is largely not subject to these assumptions, is most appropriate for detecting small changes in body composition [12,13]. However, limited accessibility and high costs often prevent the implementation of a multicomponent model. As such, DXA and BIA continue to be used as standard techniques for longitudinal body composition monitoring. Therefore, further evaluation of the ability of DXA and BIA to detect *changes* in body composition, rather than examinations of cross-sectional validity alone, is warranted.

A limited number of investigations have reported the validity of DXA or BIA for tracking changes in fat mass (FM) and FFM as compared to a multicomponent model during lifestyle interventions such as weight loss programs, preparation for athletic competition, and general exercise training [14-16]. Although these investigations provide some preliminary information regarding the relationship between body composition changes detected by DXA and BIA as compared to a 4C model, they have used substantially different participants, interventions, and assessment methodologies. Furthermore, the actual body composition changes that occurred were disparate, with FFM accretion being absent or minimal (ΔFFM: −1.1 to 0.3 kg), with one exception within a female subgroup (ΔFFM = 0.8 kg) in the study of Moon et al (2013). Although the limited amount of existing data demonstrates the need to validate DXA and BIA for assessing body composition changes during FFM accretion in general, more nuanced questions can also be raised. We propose that the validity of a body composition assessment method during FFM accretion could vary based on whether FM is simultaneously being gained or lost. BIA measures the ease with which electrical currents travel through the human body, with FM impeding the flow of current as compared to FFM [4]. Based on this principle, it stands to reason that a given change in FFM could be detected differently if FM is simultaneously being gained or lost due to the effects of FM on the bioelectrical properties of the human body and the fact that BIA FM and FFM are typically dependent on the same bioelectrical estimate. DXA imaging can estimate body composition based on comparisons of measured and theoretical ratio values, which indicate the ratio of photon attenuation at a lower energy as compared to attenuation at a higher energy [17]. Theoretically, this analysis method may allow for superior determination of body composition alterations when heterogeneous changes in FM and FFM are observed. The methodological implications of these different biophysical principles in the context of body composition assessment validity are largely unexplored. It is conceivable that scenarios in which mass from both body compartments is lost (eg, rapid weight loss) or gained (eg, overfeeding) would result in different validity of a body composition method as compared to scenarios in which one compartment increases whereas the other decreases (eg, simultaneous FFM accretion and fat loss due to lifestyle intervention or simultaneous FFM loss and fat gain due to reduced physical activity).

Based on these considerations, the purpose of the present analysis was to examine the ability of DXA and MFBIA, as compared to a rapid 4C model (ie, a 4C model produced using DXA output for bone mineral [Mo] and body volume [BV] and an independent bioimpedance technique for total body water [TBW]), to detect body composition changes in active females during FFM accretion induced by resistance training (RT) and a high-protein diet. Additionally, an examination of whether the validity of DXA and BIA varied between those who gained both FM and FFM and those who gained FFM but lost FM was performed. It was hypothesized that DXA and MFBIA would exhibit comparable utility for detecting body composition changes in the entire sample [5,16] but that the differing biophysical principles used by these methods would result in discrepancies in the subgroup analysis.

2. Methods and materials

2.1. Overview

The present secondary analysis was conducted using data previously collected from a supervised intervention in active females [18]. Participants who completed the entire intervention were included in the present analysis (n = 31; Table 1). The criterion assessment of body composition in this investigation was the rapid 4-component model [19,20]. Using body composition changes quantified by this model, 2 primary subgroups were identified within the sample:

Table 1 – Participant characteristics

	All (N = 31)		R (n = 15)		G (n = 11)		
	Means	SD	Means	SD	Means	SD	P [a]
Age (y)	22.2	2.5	22.3	2.4	21.6	1.7	.76
Body mass (kg)	62.8	7.9	63.1	7.8	61.2	9.0	.73
Height (cm)	165.9	6.6	165.2	5.0	167.1	9.3	.05
BF%	28.4	5.7	30.4	5.0	27.3	3.9	.53
FFM (kg)	44.8	4.8	43.7	3.9	44.3	5.5	.12
RT Experience (y)	5.1	2.2	5.7	2.0	4.5	2.6	.64
Current RT (d/wk)	3.1	0.7	3.1	0.8	3.1	0.6	.09
BP 1RM (kg)	40.8	10.7	40.0	10.0	36.8	6.3	.07
BP RTF (reps)	14.8	3.1	15.5	3.2	14.5	3.5	.72
LP 1RM (kg)	142.2	32.5	140.6	37.7	136.4	27.3	.41
LP RTF (reps)	13.4	6.0	14.3	6.5	11.6	6.0	.83

BP, bench press; 1RM, 1-repetition maximum; RTF, repetitions to failure; reps, repetitions; LP, leg press.
[a] From independent-samples t test between R and G subgroups.

those who gained FFM but lost FM (n = 15; "R" subgroup in the present analysis) and those who gained both FFM and FM (n = 11; "G" subgroup in the present analysis) (Figs. 1 and 2). A negligible portion of the sample lost FFM and FM (n = 3) or lost FFM and gained FM (n = 2) and were not included in the subanalysis.

2.2. *Participants and ethical approval*

Healthy females between the ages of 18 and 30 were recruited for participation. Eligibility requirements included *prior RT experience*, defined as ≥1 year of RT at a frequency of 2 to 4 sessions per week, and a BF% at screening of <34% via MFBIA (mBCA 514/515, Seca, Hamburg, Germany). Exclusion criteria were failure to meet the aforementioned eligibility requirements, current pregnancy or breastfeeding, current cigarette smoking, allergy to dairy protein, or presence of an electrical implant. This study was conducted according to the guidelines laid down in the Declaration of Helsinki, and all procedures involving human subjects/patients were approved by the Texas Tech University Institutional Review Board (IRB2017-912). Written informed consent was obtained from all participants. The original data collection was preregistered on clinicaltrials.gov (NCT03404271).

2.3. *Intervention*

All participants completed 8 weeks of supervised RT and protein supplementation. Training took place within the research laboratories under direct supervision. RT sessions were completed on 3 nonconsecutive days each week (ie Mondays, Wednesdays, and Fridays) between 12:00 PM and 6:00 PM. Upper- and lower-body sessions were alternated, with the following exercises included in the overall program at least weekly: barbell deadlift, barbell back squat, hip sled, stiff-leg deadlift, lunges with dumbbells, leg curl machine, leg extension machine, barbell bench press, bent-over dumbbell rows, barbell shoulder press, dumbbell flyes, barbell preacher curls, dumbbell triceps extensions, skullcrushers, dumbbell curls, and inverted rows. Each session included 5 to 6 exercises, with 4 sets of 8 to 12 repetitions completed for most exercises. Participants were instructed to train to

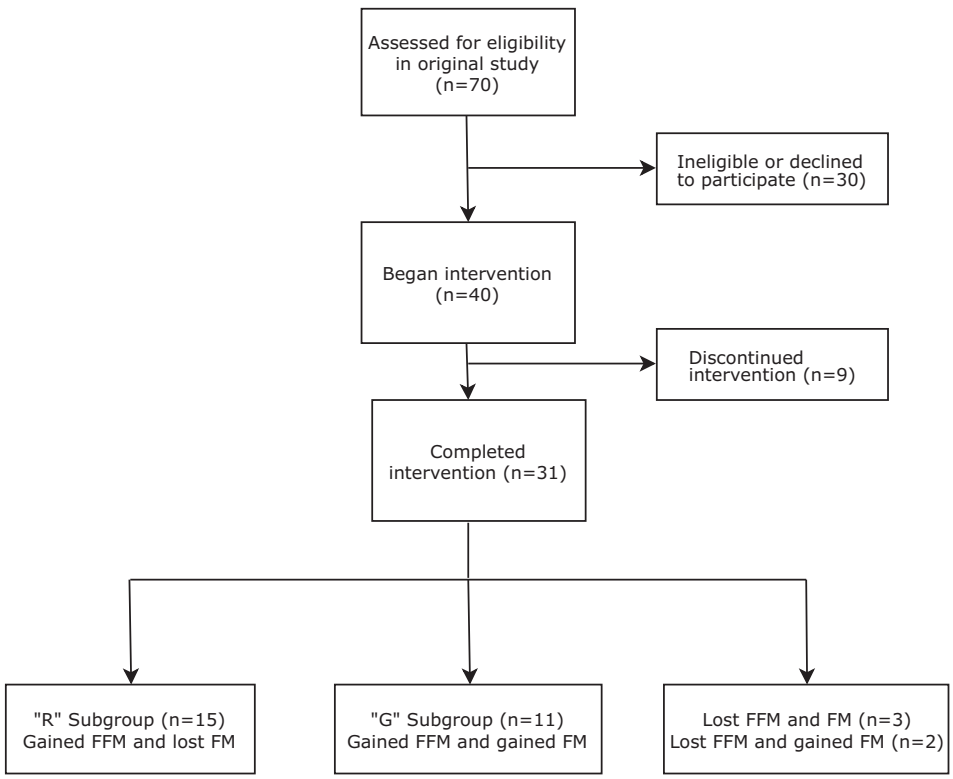

Fig. 1 – Participant selection for analysis. Participant flowchart for original data collection and present secondary analysis.

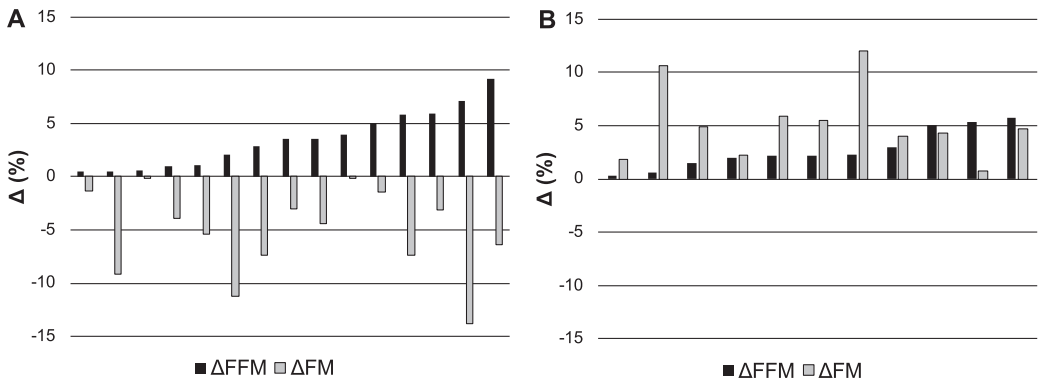

Fig. 2 – Participant subgroups based on FM changes during FFM accretion. Individual participant responses to the intervention are displayed. The majority of the total sample (N = 31) fell into 1 of 2 subgroups: (A) those who experienced a decrease in FM concurrent with FFM accretion (R subgroup; n = 15) and (B) those who experienced simultaneous increases in FM and FFM (G subgroup; n = 11).

momentary muscular exhaustion during each set, and the load was adjusted as necessary to ensure compliance with the specified repetition range. Rest intervals between sets and exercises ranged from 90 to 180 seconds. Following each RT session, participants were provided with 25 g whey protein (Elite 100% Whey, Dymatize Enterprises, LLC, Dallas, TX, USA). Participants were provided with dietary counseling and additional whey protein supplements to consume outside of the laboratory to achieve a daily protein intake of ≥1.4 g/kg [21], with actual intake being 1.6 ± 0.3 g/kg (mean ± SD). Full nutrient composition of the diet has previously been reported [18].

2.4. *Overview of laboratory assessments*

At baseline and after 8 weeks of RT, participants reported to the laboratory after abstention from eating, drinking, exercising, and consuming caffeine or nicotine for ≥8 hours. Participants were interviewed to confirm adherence to these preassessment restrictions. Participants reported to the laboratory wearing athletic clothing, and all metal and accessories were removed from the body prior to testing. Each participant voided her bladder and provided a urine sample for assessment of urine specific gravity via digital refractometer (PA201X-093, Misco, Solon, OH, USA). Participants were considered adequately hydrated if their urine specific gravity was <1.029 [22]. Each participant's body mass (BM) and height were then determined via digital scale with stadiometer (Seca 769, Hamburg, Germany).

2.5. *Body composition assessment*

The criterion body composition estimates were obtained using a rapid 4C model [19,20] produced from DXA and bioimpedance spectroscopy (BIS) output. DXA scans were performed on a Lunar Prodigy scanner (General Electric, Boston, MA, USA) with enCORE software (v. 16.2). The scanner was calibrated using a quality control block each morning

Table 2 – Cross-sectional validity of body composition estimates

	Method	Means	SD	*r*	SEE	TE	CE (CI)	EQ [a]	95% LOA	*B*	Intercept	*P* [b]
FM (kg)	4C	18.1	5.4	–	–	–	–	–	–	–	–	.45
	DXA	18.6	5.4	0.98 [c]	1.10	1.20	0.5 (−0.2, 0.9)	Y	± 2.14	−0.002	0.59	
	MFBIA	16.9	5.3	0.94 [c]	1.84	2.16	−1.2 (−0.6, −1.7)	N	± 3.62	−0.01	−0.94	
FFM (kg)	4C	44.8	4.8	–	–	–	–	–	–	–	–	.31
	DXA	44.2	4.5	0.97 [c]	1.06	1.24	−0.6 (−0.2, −0.9)	Y	± 2.21	−0.07	2.39	
	MFBIA	45.9	3.8	0.94 [c]	1.37	2.09	1.1 (0.6, 1.7)	Y	± 3.53	−0.29 [d]	11.46	
BF%	4C	28.4	5.7	–	–	–	–	–	–	–	–	.13
	DXA	29.2	5.5	0.95 [c]	1.67	1.88	0.8 (0.3, 1.4)	Y	± 3.36	−0.04	2.05	
	MFBIA	26.4	5.4	0.85 [c]	2.87	3.58	−2.0 (−1.0, −2.9)	N	± 5.99	−0.08	0.09	

Analysis based on the entire sample (N = 31).
CI, confidence interval (95% two 1e-sided confidence interval); EQ, equivalence with 4C; Y, yes (equivalent with 4C); N, no (not equivalent with 4C); *B*, regression coefficient.
[a] Equivalence with 4C based on 5% equivalence interval, which corresponds to ±0.9 kg FM, ±2.2 kg FFM, and ±1.4% body fat.
[b] *P* values from 1-way ANOVA.
[c] *P*< .0001.
[d] *P* = 0.002.

Table 3 – Validity of body composition changes

	Method	Means	SD	r [a]	SEE	TE	CE (CI)	EQ [b]	95% LOA	P [c d]
ΔFM (kg) [c]										
All (N = 31)	4C	−0.31	1.30	–	–	–	–	–	–	.90
	DXA	−0.45	1.25	0.82***	0.73	0.77	−0.13 (−0.37, 0.10)	N	± 1.51	
	MFBIA	−0.31	1.46	0.70***	1.05	1.05	0.00 (−0.32, 0.33)	N	± 2.10	
R (n = 15)	4C	−1.05	0.87	–	–	–	–	–	–	.38
	DXA	−0.83	1.08	0.73**	0.77	0.76	0.22 (−0.12, 0.56)	Y	± 1.46	
	MFBIA	−0.50	1.24	0.48†	1.12	1.21	0.55 (0.04, 1.06)	N	± 2.19	
G (n = 11)	4C	0.83	0.53	–	–	–	–		–	.35
	DXA	0.35	1.07	0.83**	0.63	0.82	−0.48 (0.09, 0.86)	N	± 1.37	
	MFBIA	0.48	0.66	0.57†	0.57	0.64	−0.35 (−0.66, −0.05)	Y	± 1.09	
ΔFFM (kg) [d]										
All (N = 31)	4C	1.07	1.16	–	–	–	–	–	–	.60
	DXA	1.32	1.31	0.71***	0.94	0.96	0.25 (−0.04, 0.54)	Y	± 1.86	
	MFBIA	1.07	0.79	0.48**	0.70	1.02	0.00 (−0.31, 0.33)	Y	± 2.03	
R (n = 15)	4C	1.52	1.12	–	–	–	–	–	–	.44
	DXA	1.41	1.46	0.75**	1.00	0.94	−0.12 (−0.57, 0.32)	Y	± 1.89	
	MFBIA	1.01	0.75	0.39	0.72	1.17	−0.51 (−0.03, 1.01)	Y	± 2.12	
G (n = 11)	4C	1.18	0.76	–	–	–	–	–	–	.23
	DXA	1.78	1.01	0.48	0.93	1.07	0.60 (0.10, 1.11)	Y	± 1.82	
	MFBIA	1.50	0.59	0.67*	0.47	0.64	0.32 (0.01, 0.64)	Y	± 1.13	
ΔBF% [e]										
All (N = 31)	4C	−0.84	1.68	–	–	–	–	–	–	.77
	DXA	−1.09	1.69	0.67***	1.27	1.36	−0.25 (−0.67, 0.16)	Y	± 2.67	
	MFBIA	−0.80	1.84	0.51**	1.60	1.71	0.04 (−0.49, 0.57)	Y	± 3.41	
R (n = 15)	4C	−1.83	1.17	–	–	–	–	–	–	.24
	DXA	−1.52	1.61	0.57*	1.37	1.33	0.31 (−0.30, 0.92)	Y	± 2.63	
	MFBIA	−0.84	1.47	0.08	1.52	1.96	0.89 (0.07, 1.71)	Y	± 3.54	
G (n = 11)	4C	0.46	0.81	–	–	–	–	–	–	.28
	DXA	−0.36	1.69	0.64*	1.37	1.51	−0.83 (−1.55, 0.10)	N	± 2.60	
	MFBIA	−0.02	0.89	0.38	0.87	1.02	−0.48 (−1.00, 0.04)	N	± 1.86	

†P < .1; *P < .05; **P < .01; ***P < .0001.
[a] No superscript on an *r* value indicates *P* > .05.
[b] Equivalence with 4C based on 100% equivalence interval, which corresponds to the 4C change observed for each variable.
[c] *P* values from 1-way ANOVA.
[d] FM and FFM values displayed in kg for mean, SD, SEE, TE, CE, and LOA.
[e] BF% values displayed in %BF for mean, SD, SEE, TE, CE, and LOA.

prior to use, and positioning of participants was conducted according to manufacturer recommendations. Each participant was able to fit within the scanning dimensions. Similar scanning techniques have produced high test-retest reliability, with technical error of measurement values of ~0.5% for lean soft tissue (LST) and ~1.5% for FM [23]. The enCORE software indicated precision error values of 0.6 kg for LST and 0.5 kg for FM, resulting in least significant change values of 1.7 kg for LST and 1.4 kg for FM. DXA bone mineral content (BMC) was divided by 0.9582 to yield an estimate Mo [24]. BV was estimated from DXA LST, FM, and BMC using the equation developed by Wilson et al (2013) for General Electric DXA scanners:

$$\mathrm{BV\ (L)} = 0.933 * \mathrm{LST} + 1.150 * \mathrm{FM} + -0.438 * \mathrm{BMC} + 1.504$$

Within our laboratory, this method of BV estimation has exhibited excellent agreement with BV estimation via air displacement plethysmography, with no significant difference between methods, statistical equivalence, no proportional bias, an R^2 value of 1.00, and a total error (TE) of 0.7 L in 179 adults [25].

BIS was used to obtain TBW estimates. BIS uses Cole modeling [26] and mixture theories [27] to predict body fluids rather than regression equations used by BIA. The BIS device used in the present study (SFB7, ImpediMed, Carlsbad, CA, USA) uses 256 measurement frequencies ranging from 4 to 1000 kHz. Each participant remained supine for ≥5 minutes immediately prior to assessment using the manufacturer-recommended hand-to-foot electrode arrangement. Duplicate assessments were performed, with the values averaged for analysis. Assessments were reviewed for quality assurance through visual inspection of Cole plots.

The 4C equation of Wang et al. [12] was used for estimation of whole-body FM:

$$\mathrm{FM\ (kg)} = 2.748 * \mathrm{BV} – 0.699 * \mathrm{TBW} + 1.129 * \mathrm{Mo} – 2.051 * \mathrm{BM}$$

FFM was calculated as BM – FM, and BF% was calculated as (FM/BM) × 100.

MFBIA was performed using a 19-frequency octapolar device (mBCA 515/514, Seca GmbH & Co, Hamburg, Germany). This device has previously been validated against a 4C model for body composition estimates and predicts FFM from bioelectrical, anthropometric, and demographic variables as previously described [28]. Previous test-retest reliability assessments in our laboratory with the specific device used in

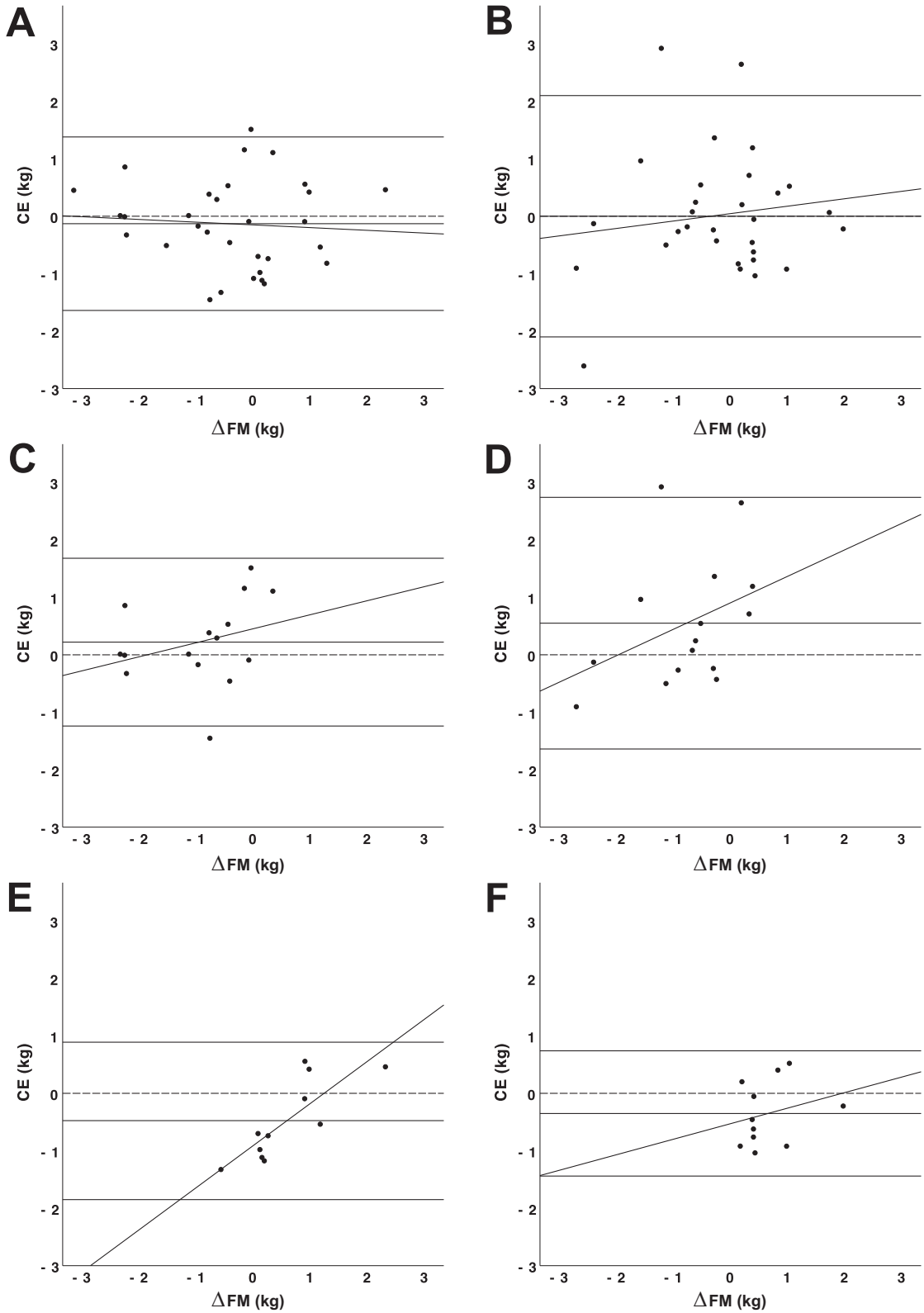

Fig. 3 – Fat mass changes. Bland-Altman plots for FM changes. Results for all participants (N = 31) are displayed in panels A (DXA) and B (MFBIA). Results for participants who decreased FM while gaining FFM (n = 15) are displayed in panels C (DXA) and D (MFBIA). Results for participants who gained both FM and FFM (n = 11) are displayed in panels E (DXA) and F (MFBIA). In each plot, the dashed horizontal line indicates zero difference between the rapid 4C model and DXA or MFBIA. The middle solid horizontal line indicates the observed CE, whereas the upper and lower solid lines represent the upper and lower limits of the LOA. The diagonal line is the linear regression line indicating the degree of proportional bias between the rapid 4C model and DXA or MFBIA. ΔFM, change in FM.

the present investigation and participant repositioning between assessments produced precision error values of 0.3%, 0.2 kg, and 0.2 kg for BF%, FFM, and FM, corresponding to least significant change values of 0.9%, 0.6 kg, and 0.7 kg for BF%, FFM, and FM, in a sample of 170 participants. Corresponding technical error of measurement values were 1.2%, 0.4%, and 1.2% for BF%, FFM, and FM.

2.6. *Statistical analyses*

The sample size for this investigation was determined via power analysis in the original trial [18]. To determine whether changes in BF%, FM, and FFM differed between 4C and alternative methods, 1-way analysis of variance (ANOVA) was performed with Bonferroni post hoc testing in the event of a statistically significant between-group difference. The constant error (CE) was calculated by subtracting the 4C value for a particular variable from the value for the alternative method, and the standard error of the estimate (SEE) was calculated using regression procedures. TE was calculated as:

$$TE = \sqrt{\Sigma(V_X - V_{4C})^2/n}$$

Where V is the value of the body composition variable in question and X is the body composition assessment method being validated (ie, DXA or MFBIA). Additionally, the Pearson product-moment correlation coefficient (r) was calculated, and strength of r values was designated as follows: 0.0 to 0.29 (negligible), 0.3 to 0.49 (low), 0.5 to 0.69 (moderate), 0.7 to 0.89 (high), and 0.9 to 1.0 (very high) [29]. The individual predictive accuracy was evaluated using the 95% limits of agreement (LOA), which were established using the methods of Bland and Altman [30]. Linear regression analysis was performed to test for proportional bias between methods [31]. Additionally, equivalence testing was performed [32]. As appropriate equivalence regions for body composition change variables have not been established, an equivalence region equal to the change detected by 4C was selected (ie, a 100% equivalence region). The rationale for this selection was that this equivalence region represented correct prediction of the directionality of the 4C change with 95% certainty. In addition to the analysis of Δ values, the aforementioned analyses were performed using baseline data to allow for reporting of cross-sectional validity of DXA and MFBIA as compared to 4C. For the cross-sectional analysis, an equivalence testing region of 5% was selected [32]. Statistical significance was accepted at $P \leq .05$. Data were analyzed using IBM SPSS (v. 25) Armonk, NY, USA, R (v. 3.6.1) Vienna, Austria, and Microsoft Excel (v. 16.11) Redmond, WA, USA.

3. Results

In the cross-sectional analysis, FM, FFM, and BF% did not differ between methods (Table 2). DXA exhibited equivalence with 4C for all variables, whereas MFBIA exhibited equivalence for FFM only. DXA exhibited very high correlations ($r \geq 0.95$), low errors, and no proportional bias for all variables. The TE and SEE values observed for DXA corresponded to the subjective rating of "ideal" [33]. MFBIA exhibited high to very high correlations ($r \geq 0.85$) and relatively low errors, but proportional bias was observed for FFM, indicating an overestimation of FFM in those with lower FFM and an underestimation of FFM in those with higher FFM. The TE and SEE values observed for MFBIA corresponded to subjective ratings of "good" to "very good" [33].

In the entire sample, ΔFM did not differ between methods (Table 3). However, neither DXA nor MFBIA exhibited equivalence with 4C for ΔFM. Conversely, both DXA and MFBIA exhibited high correlations with 4C; no proportional bias; and low CE, SEE, and TE values. Analysis in the R and G subgroups also indicated no significant differences between 4C, DXA, and MFBIA for ΔFM. However, the directionality of error metrics was disparate between R and G subgroups. Both DXA and MFBIA exhibited positive CE values (0.22 to 0.55 kg) in the R group, indicating a slight underestimation of FM loss relative to 4C, whereas both exhibited negative CE values (–0.35 to –0.48 kg) in the G group, indicating a slight underestimation of FM gain. Equivalence between 4C and DXA for ΔFM was observed in the R subgroup, and equivalence between 4C and MFBIA was observed in the G subgroup. DXA exhibited high correlations and similar SEE, TE, and LOA values for ΔFM in all 3 analyses (ie, entire sample and the R and G subgroups). In contrast, the magnitude of SEE, TE, and LOA for MFBIA was approximately twice as large in the entire sample and R subgroup as compared to the G subgroup. Additionally, although the correlation for MFBIA was high in the entire sample, correlations were low to moderate in the subgroups. The only incidence of statistically significant proportional bias for ΔFM was observed for DXA in the G subgroup (coefficient: 0.73; $P = .003$; Fig. 3).

In all participants combined, ΔFFM did not differ between methods (Table 3). Additionally, DXA and MFBIA ΔFFM exhibited equivalence with 4C in all analyses. DXA exhibited high correlations in the entire sample and the R subgroup but exhibited a low nonsignificant correlation in the G subgroup. Conversely, MFBIA exhibited a low correlation in the entire sample, a low nonsignificant correlation in the R subgroup, but a moderate correlation in the G subgroup. Both DXA and MFBIA exhibited negative CE values (–0.12 to –0.51 kg) in the R group, indicating a slight underestimation of FFM gain, whereas both exhibited positive CE values (0.32 to 0.60 kg) in the G group, indicating a slight overestimation of FFM gain. For DXA, the SEE, TE, and LOA values for ΔFFM were similar in all 3 analyses. However, the magnitude of SEE, TE, and LOA for MFBIA was approximately twice as large in the entire sample and R subgroup as compared to the G subgroup. The only incidence of statistically significant proportional bias for ΔFFM was observed for MFBIA in all participants (coefficient: –0.51; $P = .02$; Fig. 4).

In the entire sample, ΔBF% did not differ between methods (Table 3). Both DXA and MFBIA demonstrated equivalence with 4C for ΔBF% in the entire sample and R subgroup but not the G subgroup. DXA exhibited moderate correlations in all analyses, whereas MFBIA exhibited a moderate correlation for the entire sample but nonsignificant negligible to low correlations in the R and G subgroups. Similar to ΔFM, both DXA and MFBIA exhibited positive CE values (0.31% to 0.89%) for ΔBF% in the R subgroup, indicating a slight

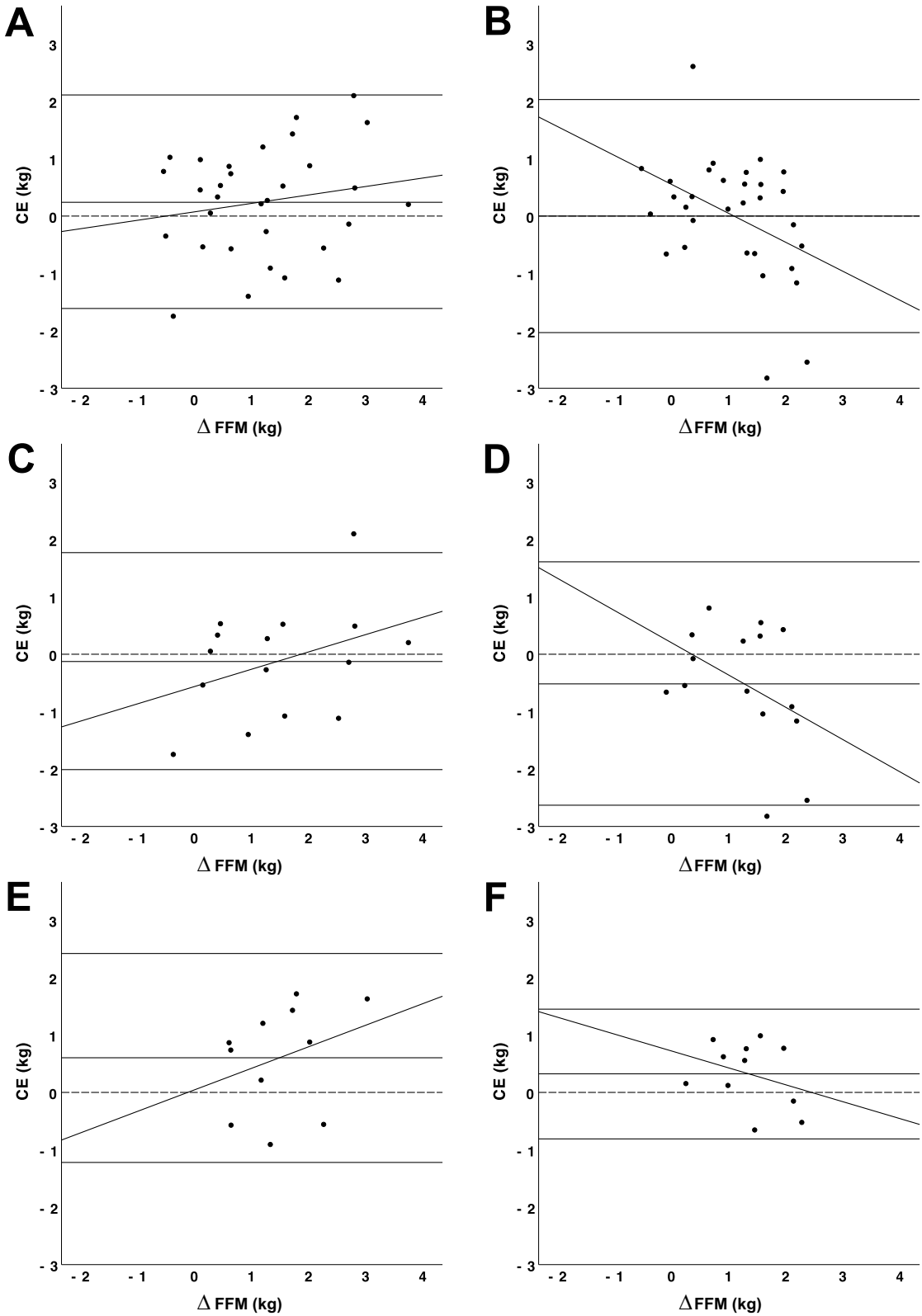

Fig. 4 – Fat-free mass changes. Bland-Altman plots for FFM changes. Results for all participants (N = 31) are displayed in panels A (DXA) and B (MFBIA). Results for participants who decreased FM while gaining FFM (n = 15) are displayed in panels C (DXA) and D (MFBIA). Results for participants who gained both FM and FFM (n = 11) are displayed in panels E (DXA) and F (MFBIA). In each plot, the dashed horizontal line indicates zero difference between the rapid 4C model and DXA or MFBIA. The middle solid horizontal line indicates the observed CE, whereas the upper and lower solid lines represent the upper and lower limits of the LOA. The diagonal line is the linear regression line indicating the degree of proportional bias between the rapid 4C model and DXA or MFBIA. ΔFFM, change in FFM.

underestimation of the decline in BF%, whereas both exhibited negative CE values (–0.48% to –0.83%) in the G group, indicating a slight underestimation of the increase in BF%. For DXA, SEE, TE, and LOA values for ΔBF% were similar in all three analyses. However, the magnitude of SEE, TE, and LOA for MFBIA was approximately twice as large in the entire sample and R subgroup as compared to the G subgroup. The only incidence of statistically significant proportional bias for ΔBF% was observed for DXA in the G subgroup (coefficient: 0.83; P = .01; Fig. 5).

4. Discussion

The purpose of this analysis was to examine the validity of DXA and MFBIA for the detection of body composition changes in active females during FFM accretion, as compared to a rapid 4C model, as well as to examine whether the validity of these methods varied depending on concurrent changes in FM. Overall, the findings of the present analysis indicate that both DXA and MFBIA may have utility for assessing body composition changes during FFM accretion, although their relative superiority may vary based on whether FM is simultaneously gained or lost. Specifically, DXA and MFBIA exhibited relatively similar performance in the entire sample, although DXA outperformed MFBIA in those who decreased FM during FFM accretion (the R subgroup), and MFBIA generally outperformed DXA in those experiencing simultaneous increases in FFM and FM (the G subgroup). Our hypothesis that DXA and MFBIA would exhibit comparable utility for detecting body composition changes in the entire sample but that differences would emerge within the subgroup analysis is generally supported by our present findings.

As expected, correlations for changes in body composition between the rapid 4C and DXA or MFBIA (r = 0.48-0.82) were weaker than the cross-sectional correlations for body composition parameters (r = 0.85-0.98) in the entire sample. Overall, DXA exhibited higher correlations with 4C body composition changes than MFBIA in all cases except ΔFFM in the G subgroup. For assessing ΔFM, DXA and MFBIA performed similarly overall in the entire sample. However, DXA exhibited a higher correlation with 4C ΔFM and lower errors in the R subgroup. In contrast, despite exhibiting a higher correlation with 4C ΔFM in the G subgroup, as well as comparable errors to MFBIA, DXA exhibited significant proportional bias. Although DXA and MFBIA exhibited similar errors for ΔFFM in the entire sample and the R subgroup, MFBIA exhibited a lower correlation with 4C and displayed proportional bias in the entire sample, unlike DXA. However, MFBIA exhibited a higher correlation with 4C ΔFFM and lower errors than DXA in the G subgroup. For ΔBF%, DXA slightly outperformed MFBIA in the entire sample and exhibited lower error in the R subgroup. However, MFBIA exhibited lower errors than DXA for ΔBF% in the G subgroup. DXA also exhibited significant proportional bias despite a higher correlation with 4C ΔBF% in the G subgroup. Differences in performance between DXA and MFBIA may be attributable to an interaction between variation in the tissue properties of FM and FFM, such as the higher water content of FFM, and the technical differences between estimation of body composition via these technologies (ie, ratio of photon attenuation vs electrical flow through body fluids and tissues) [3,4,17].

Interestingly, both DXA and MFBIA exhibited opposite directionality of group-level errors depending on whether FM was being lost or gained. Both the FM loss occurring in the R subgroup and the FM gain occurring in the G subgroup were nonsignificantly underestimated by DXA (R: 0.22 kg [21%]; G: 0.48 kg [58%]) and MFBIA (R: 0.55 kg [52%]; G: 0.35 kg [42%]). The possible underestimations of ΔFM in subgroups, which occurred regardless of the direction of ΔFM, could indicate that DXA and MFBIA are slightly less sensitive than a multicomponent model for the detection of FM changes. This interpretation aligns with previous assertions that a multicomponent model is most appropriate for detecting small changes in body composition [12,13]. The FFM gain occurring in the R subgroup was nonsignificantly underestimated by MFBIA (0.51 kg [34%]), with DXA exhibiting a negligible difference from the rapid 4C (0.12 kg [7%]). Conversely, the FFM gain occurring in the G subgroup was nonsignificantly overestimated by DXA (0.60 kg [51%]) and MFBIA (0.32 kg [27%]). These differences could indicate that the magnitude of group-level errors in estimating ΔFFM is impacted by whether FM is simultaneously gained or lost, particularly for DXA.

The merits of equivalence testing for evaluating agreement between methods have recently been detailed [32], although limited prior investigations have applied these principles to the comparison of body composition assessment methods. Based on a 5% equivalence region, DXA exhibited better cross-sectional equivalence with 4C than MFBIA. To our knowledge, no prior investigations have used equivalence testing for assessing the similarity of body composition changes over time. As such, we tentatively defined an appropriate equivalence region for detecting changes as equivalent to the change detected in the criterion method (ie, a 100% equivalence region), meaning that equivalence indicated the correct directionality of change with 95% certainty. Using this method, both DXA and MFBIA demonstrated equivalence with 4C for ΔFFM in the entire sample and both subgroups, as well as for ΔBF% in the entire sample and the R subgroup. Conversely, ΔFM changes were only equivalent between 4C and DXA for the R subgroup and equivalent between 4C and MFBIA for the G subgroup.

Few previous investigations have reported the validity of assessment methods for tracking changes in body composition as compared to a multicomponent model during lifestyle interventions [14-16]. During a weight loss program in overweight and obese women, Minderico et al (2008) reported high to very high correlations (r = 0.8-0.9) between ΔFM detected with a 4C model and multiple alternative methods, including DXA and BIA. However, negligible to moderate correlations (r = 0.1-0.5) were seen for ΔFFM. Additionally, significant group-level differences in body composition changes (ie, CE) and proportional bias were observed between 4C and alternative methods. Importantly, loss of both FM and FFM (ΔFM = –3.3 ± 3.1 kg; ΔFFM = –1.1 ± 1.5 kg) occurred at the group level. Santos et al (2010) examined the ability of DXA to track body composition changes in judo athletes preparing for competition. Changes in FM, FFM, and BF% between 4C and DXA were significantly correlated (r = 0.5-0.6), with the strength of association deemed

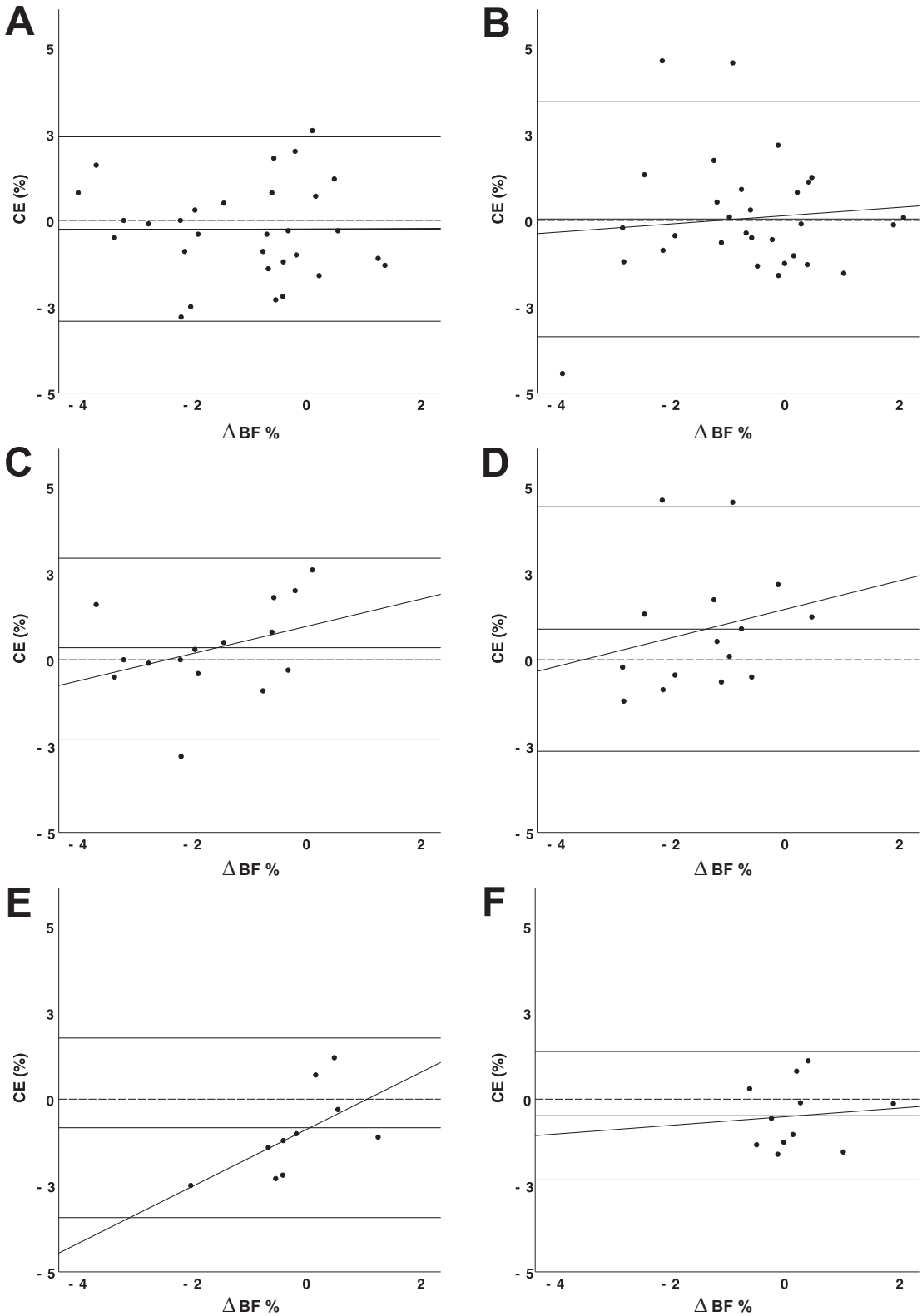

Fig. 5 – Body fat percentage changes. Bland-Altman plots for body fat percentage changes. Results for all participants (N = 31) are displayed in panels A (DXA) and B (MFBIA). Results for participants who decreased FM while gaining FFM (n = 15) are displayed in panels C (DXA) and D (MFBIA). Results for participants who gained both FM and FFM (n = 11) are displayed in panels E (DXA) and F (MFBIA). In each plot, the dashed horizontal line indicates zero difference between the rapid 4C model and DXA or MFBIA. The middle solid horizontal line indicates the observed CE, whereas the upper and lower solid lines represent the upper and lower limits of the LOA. The diagonal line is the linear regression line indicating the degree of proportional bias between the rapid 4C model and DXA or MFBIA. ΔBF%: change in body fat percentage.

moderate [29]. However, group changes in body composition, as detected by the 4C model, were small (ΔFM = –0.9 ± 2.0 kg; ΔFFM = 0.1 ± 2.0 kg). Finally, Moon et al (2013) examined the ability of DXA and multiple impedance-based equations to quantify ΔFFM in elderly males and females undergoing control conditions or an exercise intervention. In both groups, DXA exhibited negligible to low correlations (r = 0.2-0.4) for ΔFFM, whereas BIA methods varied substantially (r = 0.2-0.6). Actual body composition changes in this investigation were small (control: ΔFFM = –0.4 ± 1.2 kg in males, 0.1 ± 1.1 kg in females; exercise intervention: ΔFFM = 0.3 ± 1.3 kg in males, 0.8 ± 1.3 kg in females).

The 4C ΔFFM value (1.1 ± 1.2 kg) observed in the present study is larger than those reported in the aforementioned investigations [14-16], as well as larger than the measurement errors of the associated devices. Although most previous interventions did not induce FFM accretion (ΔFFM: –1.1 to 0.3 kg), unlike the present study, exercising female participants in the study of Moon et al (2013) exhibited relatively comparable changes (ΔFFM = 0.8 ± 1.3 kg). In this group of participants, DXA exhibited high group-level agreement with a 4C model (CE: –0.01 kg) despite a low correlation (r = 0.4) for ΔFFM. DXA also exhibited a low TE (1.28 kg) and acceptable LOA (2.53 kg), although significant proportional bias was observed (B = 0.48). In the entire sample of the present investigation, DXA exhibited a higher correlation for FFM changes (r = 0.71), lower TE (0.96 kg) and LOA (1.86 kg), and no proportional bias. Although the same model of DXA scanner was used in both investigations, the software differed, which could have influenced results. In the same investigation [16], single-frequency BIA was used, with the raw electrical data applied to 9 different FFM prediction equations. In the exercising female participants, the performance of the BIA-based equations for tracking ΔFFM varied, with r values from 0.39 to 0.64, TE from 1.10 to 1.89 kg, and LOA from 2.18 to 3.55 kg. In the entire sample of the present investigation, MFBIA performed similarly to the more accurate equations, with an r value of 0.48, a TE of 1.02 kg, and LOA of 2.03 kg. Although some similarities were present, the differences in results observed between Moon et al (2013) and the present investigation could be due to the substantial differences in average age and body composition of participants (Moon et al: 70 years, 41% body fat; present investigation: 22 years, 28% body fat).

The utilization of a rapid 4C model represents both a strength and potential limitation of the present analysis. Many body composition methodology studies simply examine the agreement between multiple noncriterion methods (eg, DXA vs BIA) rather than comparing these noncriterion methods to a criterion method (eg, a multicomponent model such as the 4C model). Within the last decade, the advent of "rapid" 4C models has presented a relative compromise between the ease of single-technique assessment methods and the strengths of a traditional 4C model. Specifically, rapid 4C models estimate body composition using 4 molecular-level components (fat, water, bone mineral, and residual) but only require laboratory assessments via DXA and BIA or BIS [19,20,34,35]. Although some assumptions are present in these models, notably the ability to accurately estimate BV from DXA output, these models have the advantage of eliminating other assumptions, such as constant FFM hydration, that are inherent to DXA and BIA as standalone techniques. Although the present study used the rapid 4C model as the criterion method, it can be considered a limitation that BV was estimated via DXA rather than air displacement plethysmography or hydrostatic weighing. However, the demonstrated validity of this BV estimation method within our laboratory adds credence to its use in the present investigation [25]. Additional limitations include the relatively limited number of participants in each subgroup and the inability to examine additional subgroups (ie, FFM and FM loss, FFM loss plus FM gain). Finally, for the subgroup analysis, participants who gained FFM were categorized dichotomously into the R or G subgroups, although some participants in these subgroups could have had nonexistent or negligible true differences in body composition changes.

In conclusion, while both DXA and MFBIA can be used to assess body composition changes during FFM accretion, they did not perform identically in the present analysis. Although DXA exhibited higher correlations with 4C for almost all body composition changes, DXA and MFBIA produced relatively comparable errors in the entire sample. However, DXA outperformed MFBIA in those who decreased FM during FFM accretion, whereas MFBIA generally outperformed DXA in those experiencing simultaneous increases in FM and FFM. Future methodological research should use nuanced evaluations of longitudinal body composition assessment validity based on the actual changes in body composition that are occurring and using rigorous analytical methods. The decision of which body composition assessment method to use in longitudinal studies should be informed by the ability of that method to accurately detect the expected changes in FM and FFM.

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